

Plan, Use Cases, Flowchart & Data Flow Diagram

A single-Admin, section-based rebuild of the container/tank readiness workflow — replacing the fixed 10-stage pipeline with a Survey-driven model where a container only moves through the sections it actually needs, every task is timed, and one Admin verifies everything. The visual design system is unchanged; this is a domain-model replacement.

Status: planning draft, derived from [SPEC.md](#) and the confirmed decisions of 2026-09-30. No schema, API, or UI code has been written yet.

Still pending from the user: the exact Survey field list, and the full container/product type-code list and its checklist mapping.

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1. Overview

ReeferReady PMS today runs on four login roles and one fixed 10-stage pipeline every container walks through identically. This plan replaces that with how the depot actually operates: one **Admin** runs the whole yard from a single dashboard; everyone else — Painter, Painter Helper, Technician, All-Rounder, Cleaner, Tea Boy — is a worker **record** Admin manages, never a login. A pre-gate-in **Survey** decides which of a fixed set of work **sections** a given container actually needs, those sections run in parallel rather than in sequence, and every task inside them is tracked with a start/stop **timer** instead of a checkbox — stopping the timer is what marks it done. Container movement (load/offload/shift) is logged separately and does not gate readiness.

The visual design system — the Crystal tokens, component styling, colour palette, the theme toggle — is explicitly unchanged. This is a data-model and workflow replacement, not a redesign.

Actors

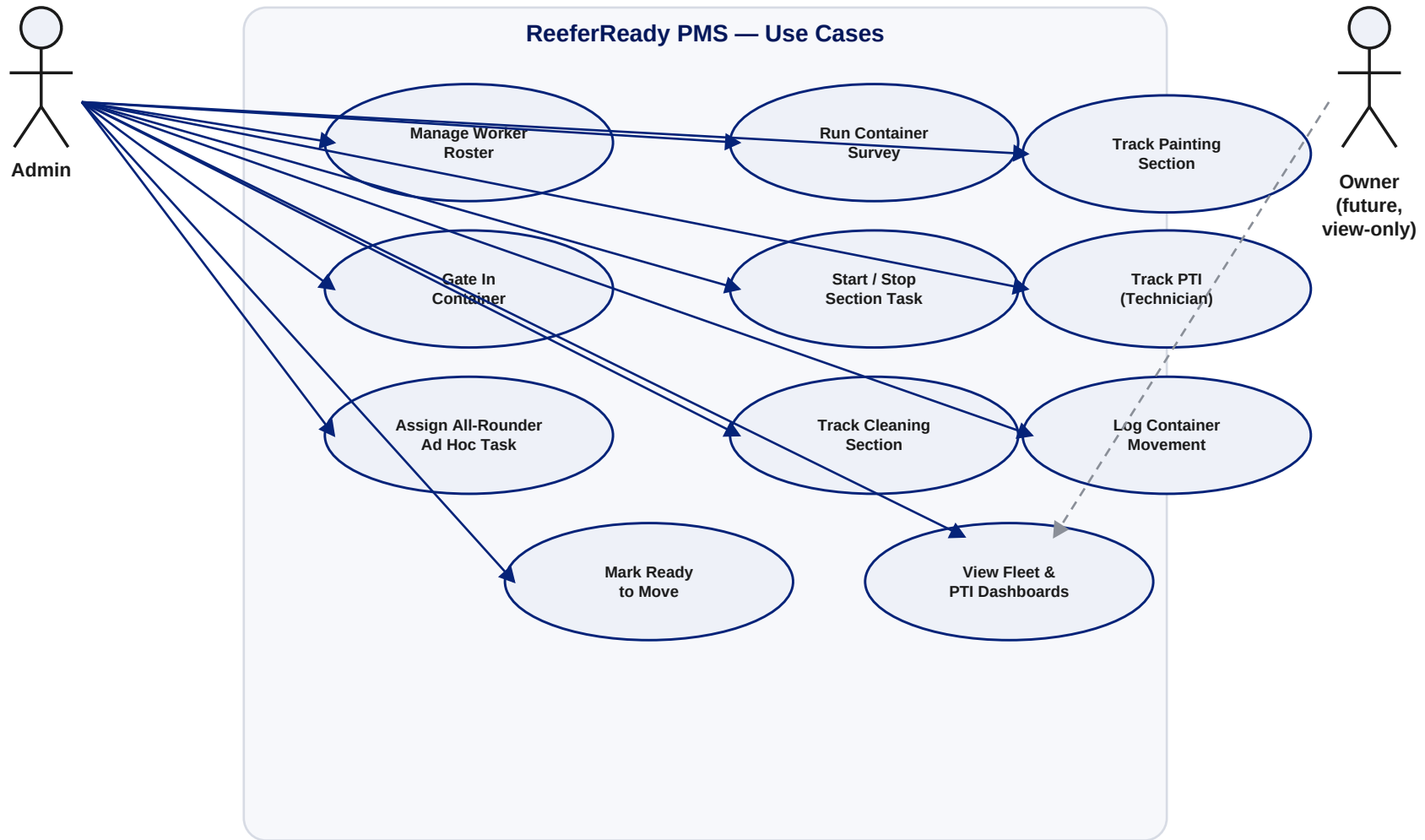
- **Admin** — the only account with a login today. Full control: manages workers, runs surveys, gates containers in, starts/stops every task timer, logs movements, views every dashboard.
- **Owner** — a possible future account. Real-time, read-only view of everything Admin sees; cannot create, edit, start/stop a timer, or log anything. Not built in the first pass, but the permission model leaves room for it.
- **Workers** (Painter, Painter Helper, Technician, All-Rounder, Cleaner, Tea Boy) — records only, never log in. Admin assigns and tracks their work on their behalf.

2. Use Case Diagram

Admin reaches every use case in the system. Owner, if built, only ever reaches the dashboard — shown dashed, since it is not part of the first build.

Use Case Diagram

Actors and the operations each can reach.



Admin performs every action; Owner (if added later) only ever reaches "View Dashboards", read-only.

3. Detailed Use Cases

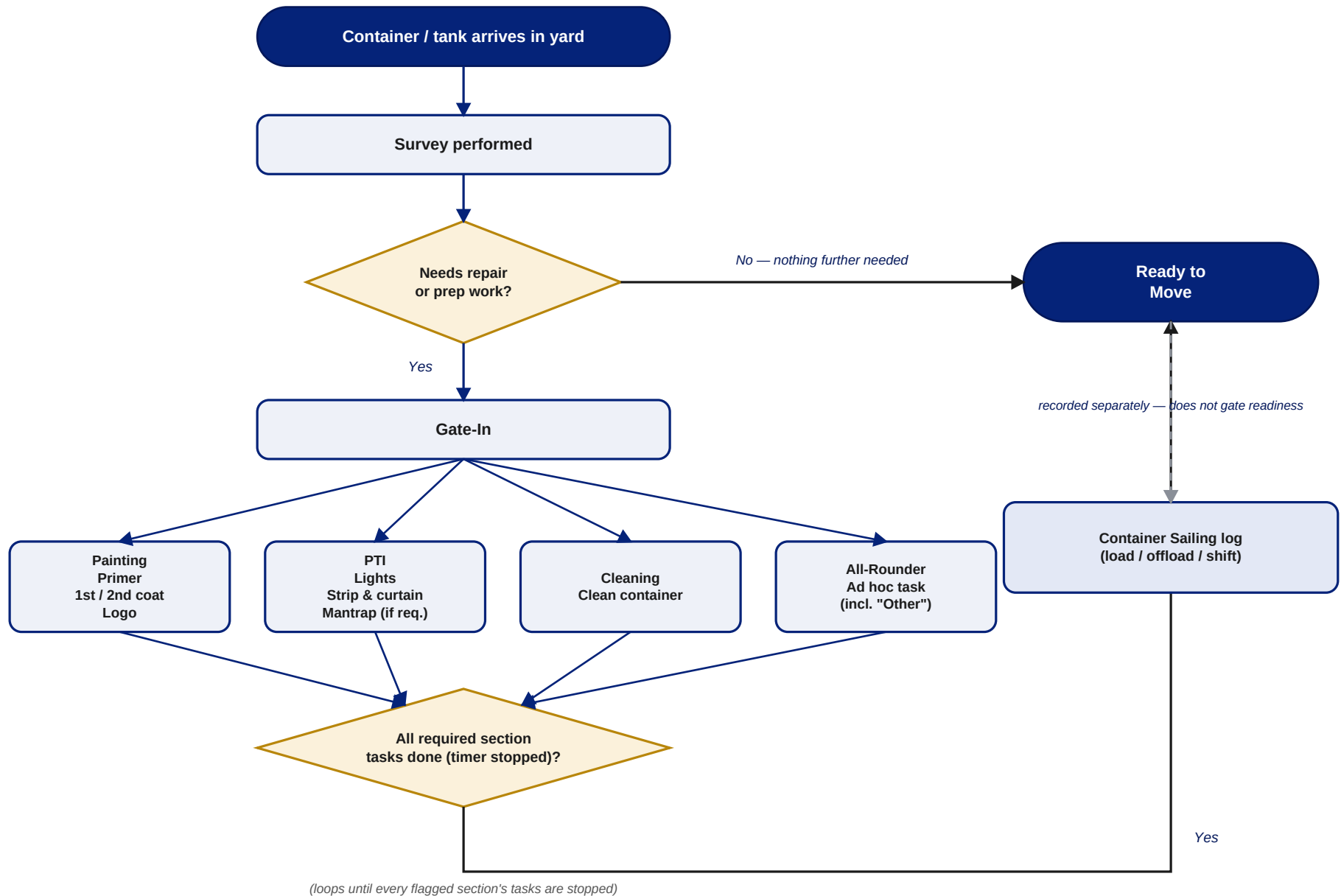
ID	Use case	Actor	Trigger	Main flow	Result
UC 1	Manage worker roster	Admin	A worker joins, leaves, or changes role	Admin opens the worker roster; adds, edits, or deactivates a record (name, type, notes).	Worker record is available for task assignment.
UC 2	Run container survey	Admin	Container/tank physically arrives in the yard, pre-gate-in	Admin records survey findings against the container; system computes the outcome.	Either flagged straight to Ready to Move, or the required sections are created.
UC 3	Gate in container	Admin	Survey outcome is "needs work"	Admin confirms arrival details — depot, customer, product/type code.	Container is active in the yard with its required sections attached.
UC 4	Start / stop a section task	Admin	A section has one or more open tasks	Admin assigns a worker to a task, starts its timer when work begins, stops it when finished (or marks it Not Applicable where the task allows it).	Task is marked done; elapsed time is recorded against it.
UC 5	Assign All-Rounder ad hoc task	Admin	A flexible worker is available and a gap needs covering	Admin picks a task type, or "Other" with a required free-text reason, and assigns an All-Rounder.	A new task is created and timed exactly like any section task.
UC 6	Log a container movement	Admin	Container is gated in and needs to be loaded, offloaded, or shifted	Admin records the event; a shift additionally records the from-site and to-site.	Movement entry is appended to the container's log — does not affect Ready to Move.
UC 7	Mark container ready to move	Admin	Every task in every required section is done	System computes the gate is satisfied; Admin confirms.	Container is flagged Ready to Move.
UC 8	View dashboards	Admin, Owner	Logged in	Open Fleet Overview, the PTI breakdown, or per-section pending counts.	Real-time operational visibility, no state changed.

4. Process Flowchart

The journey of one container from yard arrival to Ready to Move.

Container Journey — Flowchart

Survey decides which sections apply; sections run in parallel; every task is timer-driven.

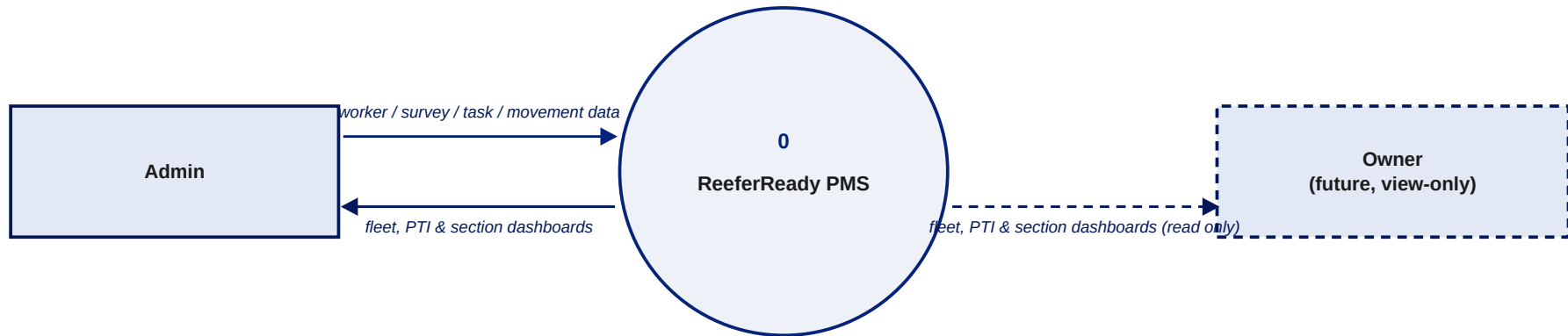


5. Data Flow Diagrams

A context diagram (Level 0) showing the system as a single process against its external actors, then a Level 1 diagram breaking it into its major processes and data stores.

DFD — Level 0 (Context)

The system as one process; Admin drives it, Owner only ever reads from it.

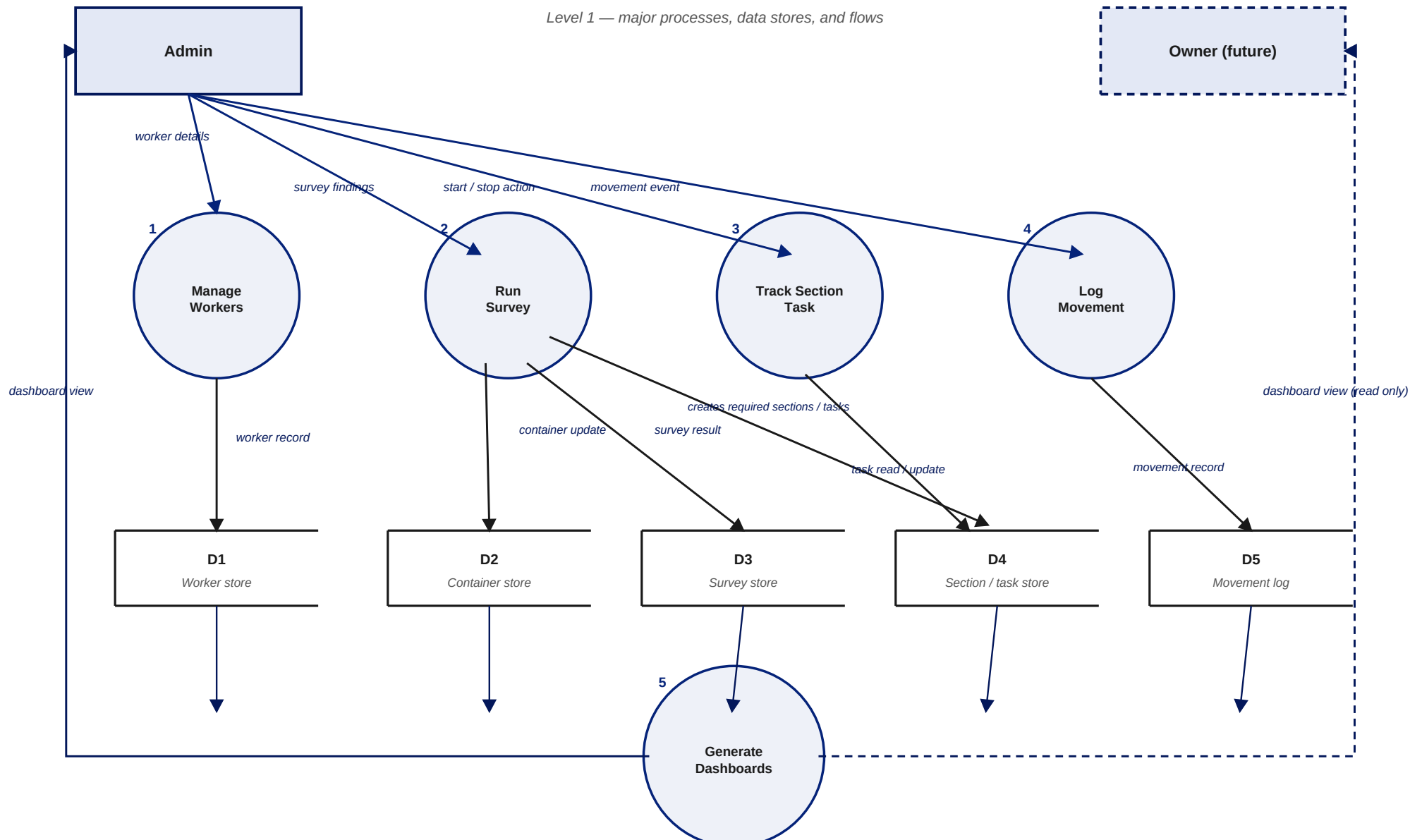


Owner sends no input — the only flow in that direction is this one, dashed, read-only view.

Level 0 — Context Diagram: the whole system as one process, its two external actors.

DFD — Level 1

Five processes, five data stores. Dashed flows are read-only lookups.



6. Data Model Summary

Full field-level detail lives in [SPEC.md](#); this is the shape at a glance.

Entity	Replaces	Holds
User	today's 4-role User	email, name, password, role (admin owner)
Worker	n/a — new	name, type (painter painter_helper technician all_rounder cleaner tea_boy sailing_crew), active, notes
Survey	n/a — new	container, findings (pending field list), outcome (ready needs-work)
ContainerSection	Container.stage	container, kind (painting pti cleaning all_rounder), required flag
SectionTask	ContainerTask	section, key, label, applicable (N/A support), assigned worker, startedAt, stoppedAt, note
MovementLogEntry	n/a — new	container, kind (load offload shift), fromSite/toSite, worker, timestamp
Container	Container (trimmed)	id, size, typeCode (pending full list), customer, depot, notes, readyAt

7. Open Items Before Implementation

- **Survey field list** — user is providing the exact fields; do not build against any earlier guess.
- **Container/product type-code list** — confirmed that type still gates the checklist, but the full list of codes and their checklist mapping is still coming.

Everything else in [SPEC.md](#)'s open-questions list is resolved as of 2026-09-30. Implementation starts once the two items above land.

<i>Planning artifact only — no schema, API, or UI code has been changed. Companion to [SPEC.md](#) in the project root.</i>